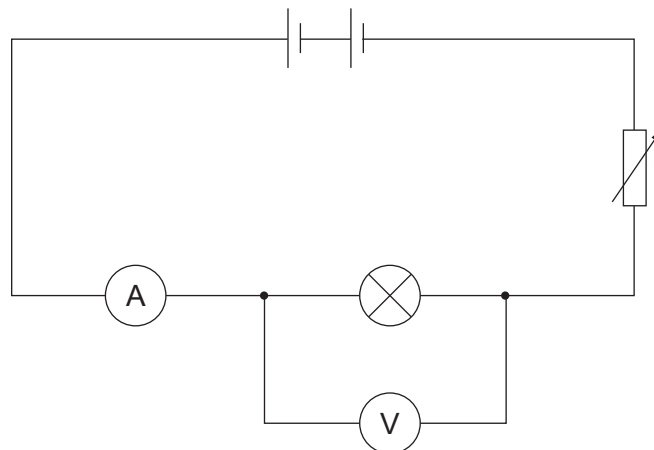


6. A student was investigating how current changes with voltage for various components. She used the following circuit to investigate a lamp.



She followed the method below.

1. Connect the circuit as shown in the diagram.
2. Set the variable resistor to give the lowest voltage and record the readings on the voltmeter and ammeter.
3. Alter the variable resistor to increase the voltage by 0.5 V.
4. Record the new readings on the voltmeter and ammeter.
5. Repeat steps 3 and 4.

(a) Some of the results are shown below.

Voltage (V)	Current (A)
0.0	0.00
2.5	0.65
3.5	0.90
4.5	1.16
5.0	1.20
5.5	1.26
6.5	1.36
8.0	1.46

(i) State the independent variable in this experiment.

[1]

.....

(ii) State the dependent variable in this experiment.

[1]

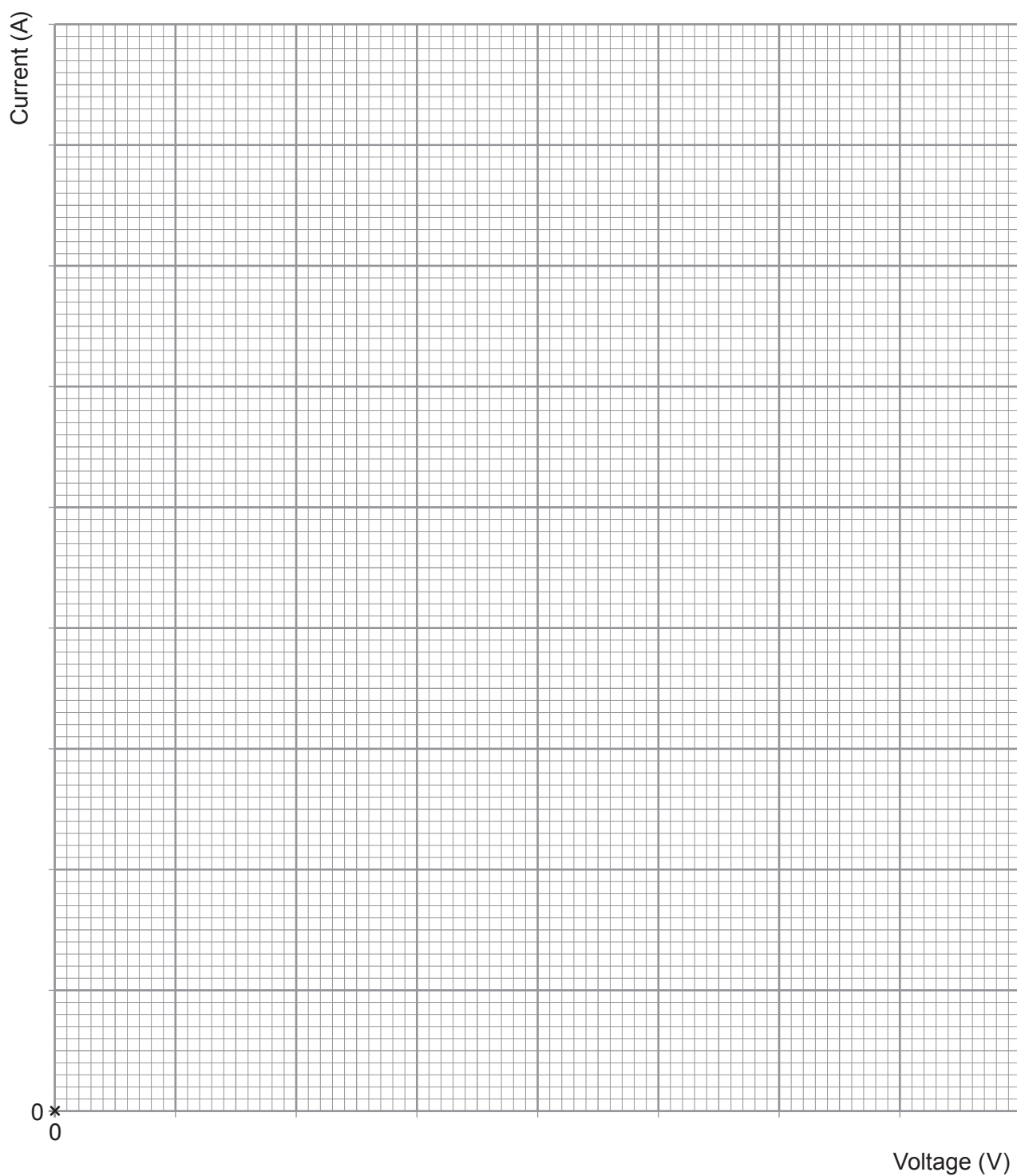
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(iii) Plot the results on the grid below and draw a suitable line.

[4]

Examiner
only



- (iv) Use your graph in (a)(iii) and the following equation

$$\text{voltage} = \text{current} \times \text{resistance}$$

to calculate the resistance of the lamp when the voltage is 7.2 V.

[3]

resistance = Ω

- (v) Explain whether the resistance of the lamp is constant throughout the experiment. [3]

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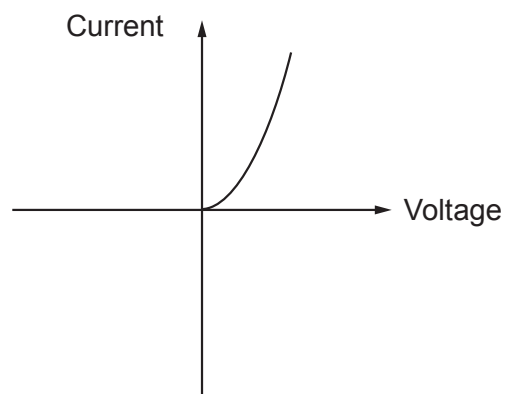
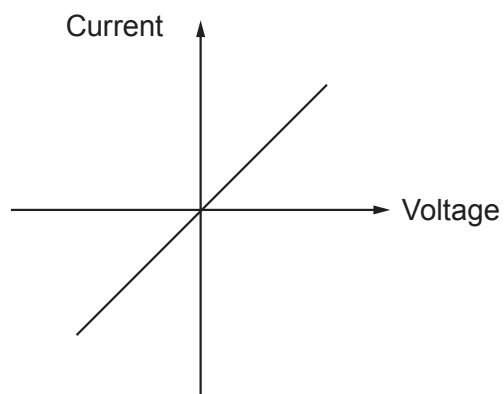
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- (b) The investigation was repeated with a resistor and then a diode. The graphs of the results are shown below.



Use the graphs to compare the variation in resistance for these components.

[4]

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END OF PAPER





THE PERIODIC TABLE

Group

1 2

3

4

5

6

7

0

¹ H Hydrogen 1

⁷ Li Lithium 3	⁹ Be Beryllium 4
²³ Na Sodium 11	²⁴ Mg Magnesium 12
³⁹ K Potassium 19	⁴⁰ Ca Calcium 20
⁸⁶ Rb Rubidium 37	⁸⁸ Sr Strontium 38
¹³³ Cs Caesium 55	¹³⁷ Ba Barium 56
²²³ Fr Francium 87	²²⁶ Ra Radium 88

¹¹ B Boron 5	¹² C Carbon 6	¹⁴ N Nitrogen 7	¹⁶ O Oxygen 8	¹⁹ F Fluorine 9	²⁰ Ne Neon 10
²⁷ Al Aluminium 13	²⁸ Si Silicon 14	³¹ P Phosphorus 15	³² S Sulfur 16	^{35.5} Cl Chlorine 17	⁴⁰ Ar Argon 18
⁷⁰ Ga Gallium 31	⁷³ Ge Germanium 32	⁷⁵ As Arsenic 33	⁷⁹ Se Selenium 34	⁸⁰ Br Bromine 35	⁸⁴ Kr Krypton 36
¹¹⁵ In Indium 49	¹¹⁹ Sn Tin 50	¹²² Sb Antimony 51	¹²⁸ Te Tellurium 52	¹²⁷ I Iodine 53	¹³¹ Xe Xenon 54
²⁰⁴ Tl Thallium 81	²⁰⁷ Pb Lead 82	²⁰⁹ Bi Bismuth 83	²¹⁰ Po Polonium 84	²¹⁰ At Astatine 85	²²² Rn Radon 86

20

Key

^{Ar} Symbol Name Z	relative atomic mass	atomic number
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